

Tier 1 Lab: EM2 - AC Circuits

AC Circuits, oscilloscopes, function generators, and impedance of capacitors and inductors.

Objectives

- Generate AC signals using a function generator.
- Measure the characteristics of AC signals using an oscilloscope
- Measure the impedance of capacitors and inductors using an oscilloscope.

Overview

This experiment will guide you through the AC signal fundamentals, its generation by function generators, and its measurement by oscilloscopes. Given its wide usage in research and teaching labs, you will explore different features of an oscilloscope and utilize it to measure the impedance of capacitors and inductors.

What to Turn In

The group as a whole will submit a single copy of the lab worksheet pdf on Gradescope, with all measurements, calculations, graphs, question answers, and signal drawings or pictures included.

Each group should acquire one dataset collaboratively. All analysis work should be equally divided and properly attributed. A separate bCourses submission is required for every Python or Excel file used in the analysis.

Theory and Background

AC and DC Circuits

In the E & M 1 lab, we built several circuits with resistors and batteries. One feature that the circuits had in common was that the current only flowed in one direction. If the electrical current only flows in one direction, such circuit is called a direct current (DC) circuit, and the source (battery in E & M1) is called a DC power supply. On the contrary, if the electrical current reverses direction periodically, the circuit is called an alternating current (AC) circuit, and the source is called an AC power supply. In this lab, we will focus on AC circuits and their measurements.

The divergence between DC and AC was much more than a technical classification in the early days of electric power transmission system. The so-called War of Currents was fought between DC-based Edison Electric Light Company and AC-based Westinghouse Electric Company to compete over the huge electric power market in the US during the late 1880s and early 1890s. As we now know, the AC system won the battle, and the reason (or the advantages of AC power in general) is its relative easiness to generate, amplify and transmit. For one thing, it is straightforward to convert a circular mechanical motion into an AC current rather than DC. Also, an AC voltage can be increased or decreased by a transformer without much loss, and thus enables efficient long-distance transmission by high voltage AC signals.

Capacitors, Inductors, and Impedance

A **capacitor** is a two-terminal device that stores electrical charge and electrical energy. It usually consists of two electrical conductors separated by a dielectric layer. The simplest kind of capacitors is the parallel-plate capacitor. If we connect its two plates to a battery, charge will flow into the plates. As the charge and voltage gradually builds up in the capacitor, current flow will decrease and eventually vanish when the voltage across the capacitor is equal to the voltage of the battery. Capacitance of a capacitor is defined as:

$$C = \frac{Q}{V},$$

where the two conductors have equal and opposite charges of magnitude Q and the voltage difference between the two terminals is V .

An **inductor** is another two-terminal device that resists the change in current and stores energy by magnetic field. The simplest kind of inductors is a metal coil. Once we connect its two ends to a battery, the current flowing through it will increase, and the resulted increasing magnetic field will induce an electromotive force. The inductance of an inductor is defined as

$$L = V / \left(\frac{dI}{dt} \right),$$

where V is the voltage difference between the two terminals/induced electromotive force (also called the EMF) and $\frac{dI}{dt}$ is the current change rate.

AC Circuits

Based on Purcell Chapter 8

Impedance

This lab concentrates on two series circuits, RC and RL, driven by an AC voltage source.

For a sine wave input, the harmonic driving term is written as $V(t) = V_o \cos(\omega t)$ where V_o is the (real) voltage amplitude and ω is the oscillation frequency. We know from experience with simple harmonic oscillators that the circuit current $I(t)$ will oscillate with the same frequency ω , but will have some phase shift with respect to $V(t)$.

Another method to solve for circuit currents is using complex exponential notation, often called “phasor” notation, which gives useful physical insight and simplifies the math. In this methodology, the input voltage and current is given by

$$\tilde{V}(t) = \tilde{V}e^{i\omega t}, \quad \tilde{I}(t) = \tilde{I}e^{i\omega t},$$

where the tilde indicates a complex quantity, consistent with Purcell's notation. As a reminder, Euler's formula states that $e^{ix} = \cos(x) + i \sin(x)$, where $i = \sqrt{-1}$.

In AC circuits, the role of resistance is replaced by the closely-related term, **impedance**, defined as

$$Z = \frac{\tilde{V}(t)}{\tilde{I}(t)} = \frac{\tilde{V}e^{i\omega t}}{\tilde{I}e^{i\omega t}} = \frac{\tilde{V}}{\tilde{I}},$$

where Z is generally a complex quantity with units of resistance.

Impedance of a Resistor

For a resistor, the voltage and is related to the current through Ohm's Law $\tilde{V}(t) = R\tilde{I}(t)$. Using the definition of impedance $Z = \tilde{V}/\tilde{I}$, we get that the **complex impedance of an resistor** is actually real and is equal to the resistance, $Z_R = R$.

Impedance of an Inductor

For an inductor, we know that

$$\tilde{V}(t) = L \frac{d\tilde{I}(t)}{dt}.$$

Plugging in our expressions for the complex voltage and current gives $\tilde{V} = i\omega L\tilde{I}$. Using the definition of impedance $Z = \tilde{V}/\tilde{I}$, we get that the **complex impedance of an inductor** is $Z_L = i\omega L$.

Impedance of a Capacitor

For a capacitor, we know that

$$\tilde{I}(t) = \frac{d\tilde{Q}(t)}{dt} = C \frac{d\tilde{V}(t)}{dt}.$$

Plugging in our expressions for the complex voltage and current gives $\tilde{V} = \tilde{I}/i\omega C$. Using the definition of impedance $Z = \tilde{V}/\tilde{I}$, we get that the **complex impedance of a capacitor** is $Z_C = \frac{1}{i\omega C}$.

Solving AC Circuits

We can solve for the currents and voltage of an AC circuit using the complex impedances and the AC Kirchhoff's Laws. The generalized series circuit shown in Figure 1 has two elements, Z_1 and Z_2 , and an input voltage $\tilde{V}(t) = \tilde{V}e^{i\omega t}$.

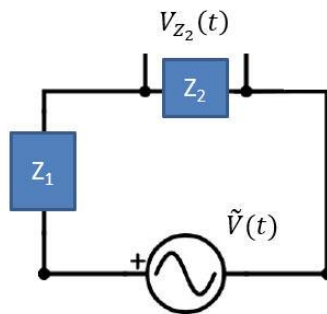


Figure 1. A generalized AC circuit.

The circuit current $\tilde{I}(t) = \tilde{I}e^{i\omega t}$ can be calculated using the AC Kirchhoff's Law:

$$\tilde{V}e^{i\omega t} = \tilde{I}e^{i\omega t}Z,$$

where $Z = Z_1 + Z_2$.

The time-dependent output voltage across the circuit element Z_2 in Figure 1 is given by:

$$\tilde{V}_{Z_2}(t) = Z_2 \cdot \tilde{I}(t) = Z_2 \cdot \frac{\tilde{V}e^{i\omega t}}{Z}. \quad (1)$$

Equation (1) can also be written in terms of the amplitude and phase as

$$\tilde{V}_{Z_2}(t) = |\tilde{V}_{Z_2}| e^{i\phi_{Z_2}} e^{i\omega t} = |\tilde{V}_{Z_2}| \exp(i(\omega t + \phi_{Z_2})),$$

where $|\tilde{V}_{Z_2}| = \sqrt{\tilde{V}_{Z_2}^* \tilde{V}_{Z_2}}$ and the phase angle across the specified circuit element relative to the driving voltage is given by

$$\phi_{Z_2} = \tan^{-1} \left(\frac{\text{Im}\{\tilde{V}_{Z_2}\}}{\text{Re}\{\tilde{V}_{Z_2}\}} \right).$$

Even though calculations are elegantly handled by complex numbers, we can only measure *real* quantities with our lab equipment! Therefore, the quantities of interest for this lab is the real part of the complex voltage. For example, the voltage V_{Z_2} that is physically measured will be

$$V_{Z_2} = \text{Re}\{\tilde{V}_{Z_2}(t)\} = |\tilde{V}_{Z_2}| \cos(\omega t + \phi_{Z_2}).$$

Because the phase measurements are all relative, the complex voltage $\tilde{V}$ of the AC voltage *source* is generally assigned to be a real quantity, or $\tilde{V} = V_o$.

RC Circuit – Low-Pass Filter

For an RC series circuit with element 1 being the resistor and element 2 being the capacitor, $Z_1 = R$ and $Z_2 = \frac{1}{i\omega C}$. The complex voltage across the capacitor is then

$$\tilde{V}_{Z_2}(t) = \tilde{V}_C(t) = \frac{1}{i\omega C} \cdot \frac{V_o e^{i\omega t}}{R + \frac{1}{i\omega C}} = \frac{V_o e^{i\omega t}}{1 + i\omega RC} = \tilde{V}_C e^{i\omega t}. \quad (2)$$

The real quantities of interest are:

$$V_C(t) = \frac{V_o}{\sqrt{1 + (\omega RC)^2}} \cos(\omega t + \phi_C) = V_C \cos(\omega t + \phi_C), \quad (2a)$$

$$\phi_C = \tan^{-1}(-\omega RC). \quad (2b)$$

This configuration is known as a “low-pass” filter and its frequency response is shown below in Figure 2.

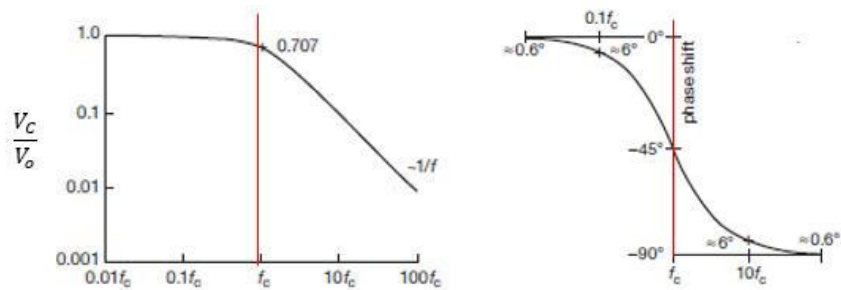


Figure 2 Frequency response (amplitude and phase) of a low-pass RC circuit, plotted on logarithmic axes, where f_c^{RC} is the cutoff frequency. From Horowitz and Hill, *The Art of Electronics*, 3rd edition.

LR Circuit – High-Pass Filter

For an LR series circuit with element 1 being the resistor and element 2 being the inductor, $Z_1 = R$ and $Z_2 = i\omega L$. The complex voltage across the inductor is then

$$\tilde{V}_{Z_2}(t) = \tilde{V}_L(t) = i\omega L \cdot \frac{V_o e^{i\omega t}}{R + i\omega L} = \tilde{V}_L e^{i\omega t}. \quad (3)$$

The real quantities of interest are:

$$V_L(t) = \frac{V_0 \omega L}{\sqrt{R^2 + (\omega L)^2}} \cos(\omega t + \phi_L) = V_L \cos(\omega t + \phi_L), \quad (3a)$$

$$\phi_L = \tan^{-1}\left(\frac{R}{\omega L}\right). \quad (3b)$$

This configuration is known as a “high-pass” filter and its frequency response is shown below in Figure 3.

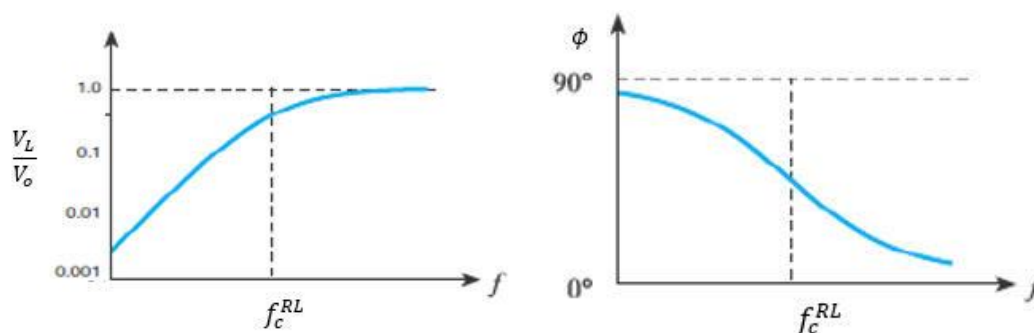


Figure 3. Frequency response (amplitude and phase) of a high-pass RL circuit, plotted on logarithmic axes, where f_c^{RL} is the cutoff frequency.

Equipment and Equipment Notes

- MSO 2024B Oscilloscope
- Tektronix AFG2021 Function Generator
- Breadboard
- Wires, Cables, and Adapters
 - Jumper Wires (pins at both ends)
 - BNC-to-BNC Coaxial Cables
 - BNC-to-Minigrabber Coaxial Cables
 - “T” BNC Adapter
- Circuit Elements
 - Resistors: 1 k Ω , 10 k $\Omega \pm 1\%$
 - Capacitor: 1 $\mu\text{F} \pm 5\%$
 - Inductor: 100 mH $\pm 10\%$
- Resistors: 1 Ω (x3), 1 k Ω (x1), 4.7 k Ω (x2), 10 k Ω (x2)

Equipment Notes - Oscilloscopes

Measurement Principles

Oscilloscopes are the most widely used tool to measure time-dependent electrical signals. It is important to keep in mind that oscilloscopes can only measure voltage (not current, or resistance), and thus...

An oscilloscope is simply a high-end multi-channel voltmeter with a very high acquisition rate, very good resolution, and additional plotting abilities.

Oscilloscopes also measure the period and frequency of the signal. The measurement principle of an analog oscilloscope is shown in Figure 4 below. In 5BL we will exclusively use digital oscilloscope in this lab. However, in many research labs (including the 111 series labs) you might come across analog oscilloscopes, so it is good to know a little bit about their operation.

Just like an old-fashioned television set, an analog oscilloscope utilizes an electron beam to create patterns on a screen. The electron beam is generated and accelerated in an electron gun (usually a CRT - cathode ray tube) and deflected by both vertical and horizontal voltages before shooting onto a phosphor screen to emit light which is the signal that

we see. All the tricks that an oscilloscope can do ultimately trace back to the tuning of the vertical and horizontal voltages. Let's try to understand more about how this voltage control works.

Let's assume we are measuring a sine wave (voltage) using the oscilloscope. As shown in Figure 4, what an oscilloscope actually does is to apply this input voltage (after amplification) to the vertical plates. If nothing is applied to the horizontal plates, you will only see a line along the y direction on the screen. In order to "spread over" the sine wave to make it visually accessible, some sort of horizontal scanning will be required. For common oscilloscopes, they utilize a sawtooth waveform as shown in Figure 4 to fulfill this purpose: at $t=0$, it will deflect the electrons to the leftmost position, and as time passes, the electron beams will be shifted rightward linearly with time. In this way, we will see a sine wave as shown on the screen in the Figure.

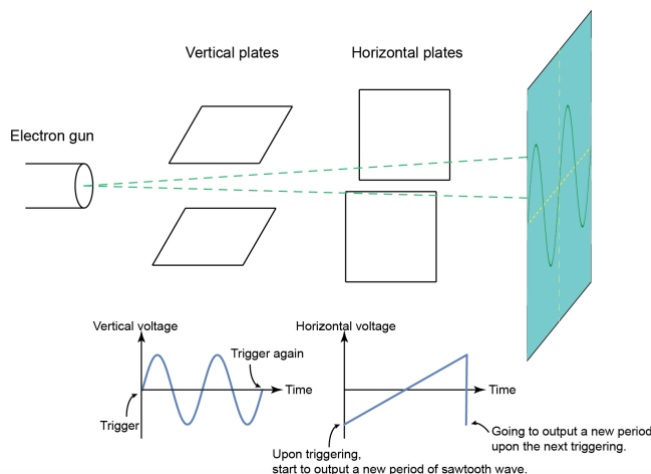


Figure 4. Principles of an analog oscilloscope. An electron beam, produced by an electron gun, experiences both vertical and horizontal bending before shooting onto a phosphor screen to emit light. Your input signal (sine wave here) will be amplified and supplied to the vertical plates, and moves the electron beam up and down, while a horizontal sawtooth wave (automatically generated, or "triggered") will move the electron beam left and right to help display the vertical voltage waveform.

The MSO 2024B Oscilloscope

With measurement principle in mind, let's have a look at the real oscilloscope we are going to use, the MSO 2024B (MSO stands for **Mixed Signal Oscilloscope**), shown in Figure 5. It is a digital oscilloscope, and the waveform is displayed on an LCD screen unlike analog ones that use CRTs. There are **4 analog input channels** and **16 digital input channels**. The analog channels are numbered from 1-4. To select any channel to be displayed on the screen, you simply press the corresponding colored number buttons on the panel in vertical control area (shown in Figure 6 as "Channel Buttons"). In this lab we only input the signals through analog channels. The panel can be divided into four major parts: **display**, **vertical control**, **horizontal control** and **trigger control**. There are many buttons on the panel but in this lab, we will only focus on the most important features in each part for this lab.

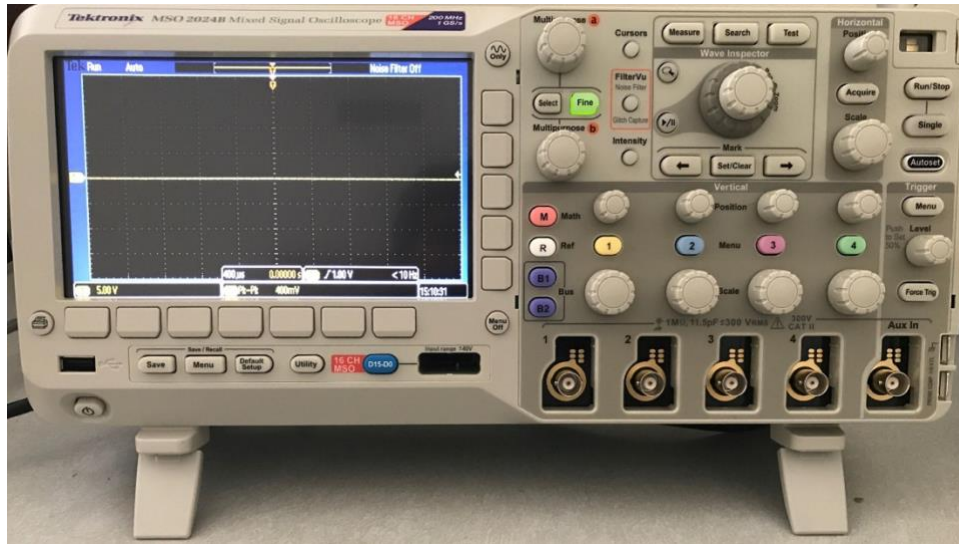


Figure 5. Front panel of our oscilloscope (Tektronix MSO 2024B). It is divided into five major regions: display area, vertical control, horizontal control, trigger control and wave inspector area.

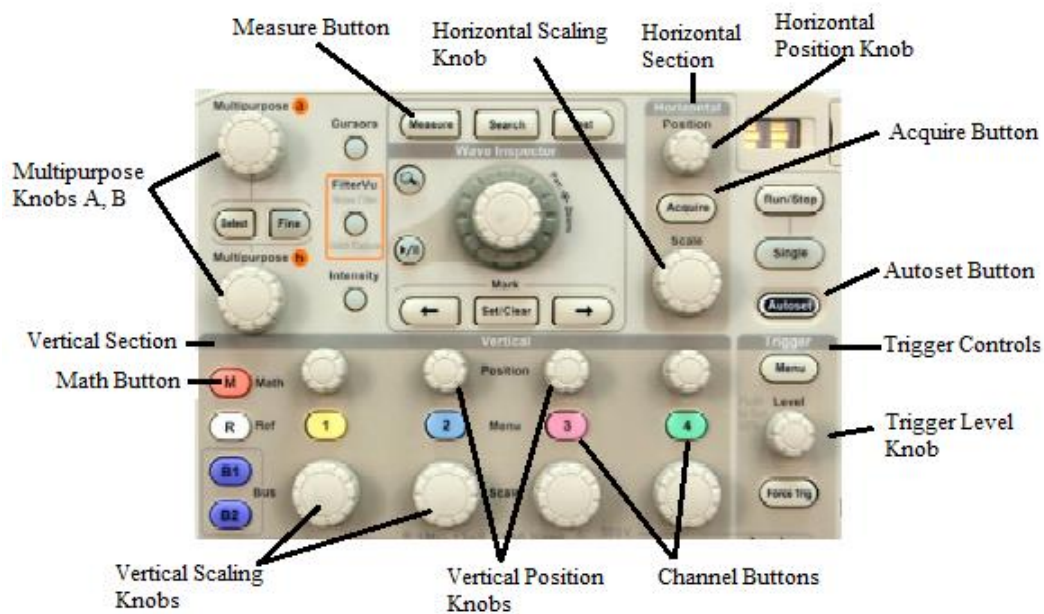


Figure 6. The buttons in the front panel of the oscilloscope (Tektronix MSO 2024B) and their function.

The four most important controls, which will be discussed in more detail later, are:

1. The **Vertical Scaling Knobs** (bottom center) adjust the vertical size of the image on the scope screen; a typical setting would be 1 V/div (which sets the vertical scale to “1 volt-per-division”).
2. The **Vertical Position Knobs** (above the vertical scaling knobs) move the scope traces up and down.
3. The **Horizontal Scaling Knob** (the large knob second down in the second column from the right) adjusts how fast the scope draws the image across the screen; a typical setting would be 1 ms/div (which sets the horizontal scale to “1 millisecond per division”).

4. The **Trigger Level Knob** (bottom right) adjusts the triggering voltage.

The signal itself will be connected to one of the input BNC jacks on the scope as shown in the picture on the right.



Our scopes have four channels and can display up to four signals simultaneously.

Viewing an Electronic Signal

Using a BNC cable, connect the signal from the output of **Tektronix AFG 2021 Signal Generator** (the “Channel Output” one labeled 50 Ω) signal generator to channel 1 on the scope. The scope channels are located in the front on the bottom right of the scope and are labeled “1, 2, 3, and 4”.

Turn on the signal generator and you will see the default sine wave on the display of the signal generator. Push the “Channel On/Off” button of the signal generator to the “On” position and the green light will come on (refer to Figure 7 to connect your signal generator to the oscilloscope). You should now see the signal on the oscilloscope as well.



Figure 7. A Signal generator connected to the oscilloscope. Note the 3 illuminated buttons on the signal generator.

Vertical and Horizontal Scaling and Position Controls

The vertical scale for each of the scope's channels is set with the knobs labeled “Scale” in the Vertical section of the scope controls and has units of volts/division. The horizontal scale for all of the channels is set with the Scale knob in the Horizontal section of the scope controls and has units of seconds/division. A division is the scope distance between the dashed lines, with tick marks in the center axes. As an example, a 100Hz 10 volt peak to peak triangle wave (or sine wave) will take up two divisions on the 5V/division scale and one period will take up 10 divisions on the 1 ms/division scale. The current value of the vertical and horizontal scales is displayed at the bottom of the scope's screen.

You can also adjust the position of the signal on the screen with the position controls. These are the smaller sized knobs located in the Vertical and Horizontal sections of the scope controls. The solid arrows on the left side of the screen indicate where the ground level for each channel is.

The Scale (time-base) knob is the primary horizontal control and controls the rate at which the scope trace sweeps (from left to right) across the screen. A scope trace of a 1 μ s period sine wave (1MHz) displayed on the 200 ns/div scale, would show two complete cycles of the wave.



Figure 8. The scope trigger menu

Look at the scope's "Trigger" section. The trigger circuit determines when the scope starts its horizontal display sweep, but it needs an input. The main purpose for triggering a signal is to display a desired signal with minimum interference from other sources. Press the "Menu" button under the Trigger area. You will see another menu displayed on the screen (Figure 8). Click on the button underneath the "Type" menu item. A drop-down menu opens with the "Edge" option highlighted. This means that the triggering function works at the edge of the signal. This might not mean anything to you now; however, as you get more familiar with the oscilloscope, you will learn more about different features available to you. Use the "Multipurpose A" knob to navigate vertically through the menu. To exit the Trigger menu, press the Menu button in the Trigger area or the "Menu Off" button near the scope display. You don't need to worry about different triggering your signal now, and we will revisit this function in later labs.

One control deserves special mention for triggering: the **Autoset** button. If you cannot find the scope trace, push this button. The trace will be displayed onto the screen with reasonable voltage and time settings. Some settings, such as the 10X probe or coupling, might need adjustment.

A very useful feature of these oscilloscope is the **cursor**. The cursor allows you to make measurements of time and voltage between two points (or levels). Figure 9 shows where the cursor button is located on the oscilloscope. Press the "Cursors" button. You should see two vertical lines on the scope face. If you press the button again, the vertical lines will become horizontal ones. Press it one last time to see both horizontal and vertical lines, as depicted in Figure 10.



Figure 9. The cursors button.

The vertical lines are used to determine the time difference between the two cursor lines, and the horizontal lines will give you the voltage difference between the horizontal cursor lines. You can only control one set of lines at a time. The solid line is the line that you can control by pressing the "Select" button underneath Multipurpose a button (see Figures 9 and 10). If you press the button again, the other set of lines become solid. Use Multipurpose dials "a" or "b" (see Figures 9 and 10) to move these solid lines to the right or left or up or down, depending on which set of

cursors you have invoked. The “Fine” button toggles between coarse and fine adjustments with the vertical and horizontal position knobs, the trigger level knob, and the operations of multipurpose knobs **a** and **b**.

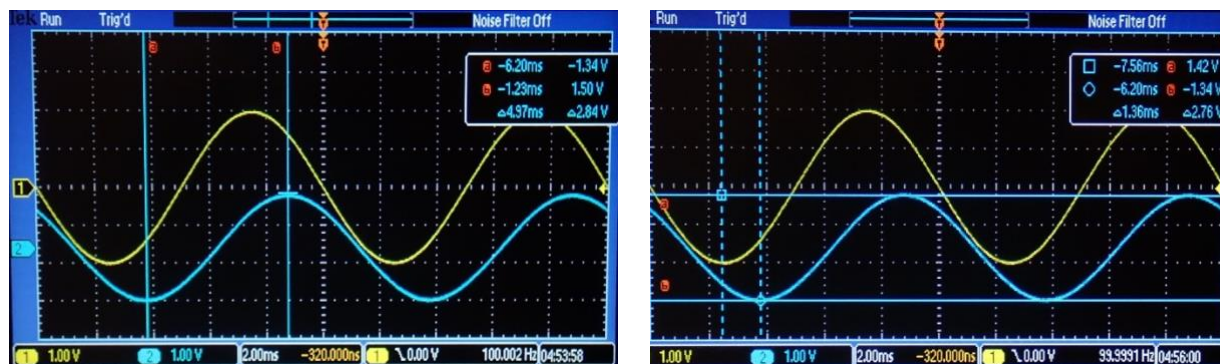


Figure 10 This figure displays the two cursor configurations, the left with two cursors and right one with four. The display box on the top right shows the numerical value of the positions of the cursors on the first two lines. The third line displays Δt and ΔV .

The values of these two cursors can be recorded using the information box on the top right of the display screen on the oscilloscope.

You will see oscilloscopes quite often in the future. If you are interested in getting more prepared, you can read the oscilloscope manual located in the 5BL bCourses files.

Part 0: Procedure and Data Collection

This part outlines the procedures and data collection you should be following in the experiments that follow and gives your preliminary practice to familiarize yourself with the oscilloscope and function generators. You can find supplemental information on videos posted to the course site:

- EM2 Procedure and Data Collection
- EM2 Oscilloscope Operation

Practice Measuring Using an Oscilloscope

Setup of a Function Generator and an Oscilloscope

- Turn on the function generator
- Connect the output port of your function generator to CH1 of your oscilloscope using a BNC coaxial cable, as shown in Figure 5.
- Select the sine waveform from the function generator bottom left buttons.
- Change the frequency to 1 kHz: Press the button next to the “Frequency/Period/Phase Menu” option displayed on the screen, press the “Frequency” button, input “1”, then press on the button next to where “kHz” is displayed. Your signal generator should be producing a 1kHz sine wave.
- Use the up-arrow key to return to the main menu
- Set the load impedance: press the button next to the “Output Menu” option for the sub-menu, then press the button next to “Load Impedance” for the sub-menu, then select the “Hi-Z” button in the next menu. The signal generator output impedance is now properly aligned with the scope input impedance. Go to the main menu.
- Set the waveform amplitude: from the main menu, press the button next to the “Amplitude/Level Menu”, press the “Amplitude” button, input “4”, then press “ V_{pp} ”.
- Deliver the signal to the oscilloscope: press the green “on/off” button on your signal generator. This is NOT the same as the on/off power button.
- For the oscilloscope, in the vertical control panel, press “1” to select CH1 (yellow trace). When the control menu opens, make sure the “DC Coupling” is chosen and the “Invert” option is off. In addition, the “Probe Setup” should be set to “1×”. To change the value, if needed, just press the button under the menu item and choose “1×”.
- Press the “Menu Off” button near the screen to turn off the displayed scope channel horizontal menu.
- Select the volts/div to be 1 by turning the vertical channel 1 “Scale” knob to that value.
- In the horizontal control panel, set the sec/div to be 400 μ s by turning the “Scale” knob.
- In the Trigger control panel menu, set the trigger source to be CH1, the trigger mode to be Auto, the trigger level to roughly in the middle range and the slope to be positive (rising edge).

Learn to Adjust the Oscilloscope

- Now your task is to display a nice stable 1 kHz 4V peak-to-peak (V_{pp}) sine wave on the oscilloscope. You may need to play with the scope knobs several times to achieve it.
- If the displayed signal is shifting rapidly, the oscilloscope is not triggered (which you can also tell if the **Trig’d** in the top left is not showing). Adjust the **Level** knob in the trigger control to fit it in the signal range until you can see a stable signal or press **Autoset** if all else fails.
- If you want, you can adjust the vertical or horizontal position knobs on your oscilloscope to better position the signal on the display.

- If your V_{pp} bigger or smaller than the $4V_{pp}$, adjust the amplitude of the function generator.
- If the period is bigger or smaller than 1ms ($f = 1$ kHz), fine adjust using the frequency knob of your function generator.
- Now, you should have a desired 1kHz 4V peak-to-peak (V_{pp}) sine wave. Count the number of divisions (also referred to as div), both major and minor, and convert it to volt or second by your setting parameters.

IMPORTANT: The channel input consists of an amplifier followed by an analog-digital converter (ADC). The scope displays the ADC output, which has a fixed number of bits, filling the scope face amplitude divisions. The most accurate amplitude measurements are made when the signal fills, but not overfills, the scope display so the signal uses ADC bits most efficiently. If needed, change the channel amplitude setting during your measurements, keeping track of the signal on both scales.

- Take a picture of the scope face OR draw your signal below, for the lab submission.

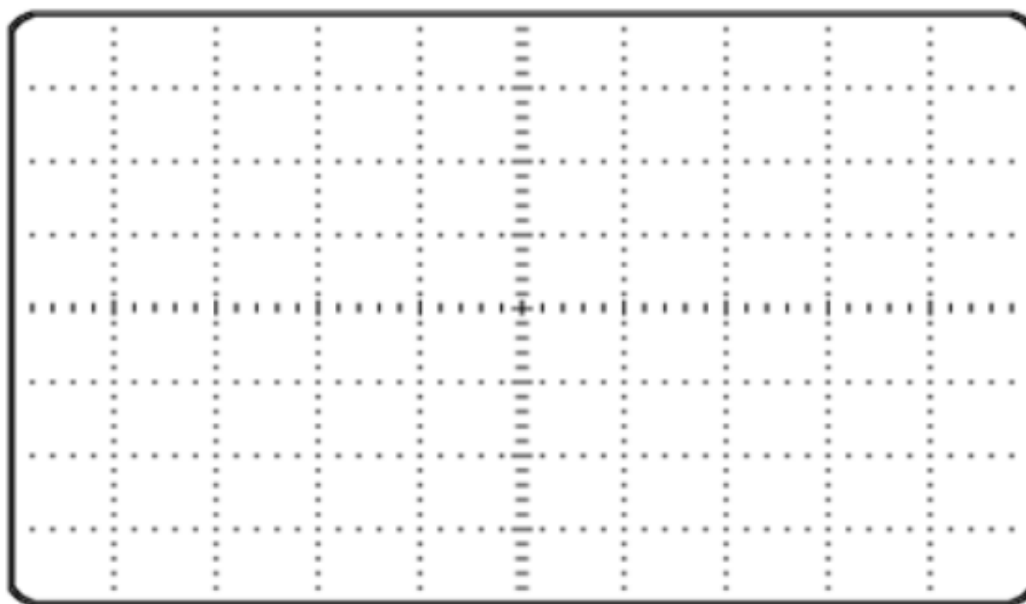


Figure 11. Oscilloscope Face for Part 0.

- Fill in the following formula, from the waveform:

$$\begin{aligned} \underline{4} \text{ div} \times 1 \text{ volts/div} &= \underline{4} V_{pp} \\ \underline{3.7} \text{ div} \times 400 \mu\text{s/div} &= \underline{\quad} \text{msec} \end{aligned}$$

Example: Major and minor divisions are shown on Figure 12. For the vertical scale, one div is 1.0V (1.00 V in the Figure) while for the horizontal time scale, one div is $400 \mu\text{s}$ ($400 \mu\text{s}$ in the Figure). There are 5 minor divisions in 1 major division. In the Figure, the V_{pp} reading is 4 major + no minor division in the vertical scale, $V_{pp} = 4.0V$.

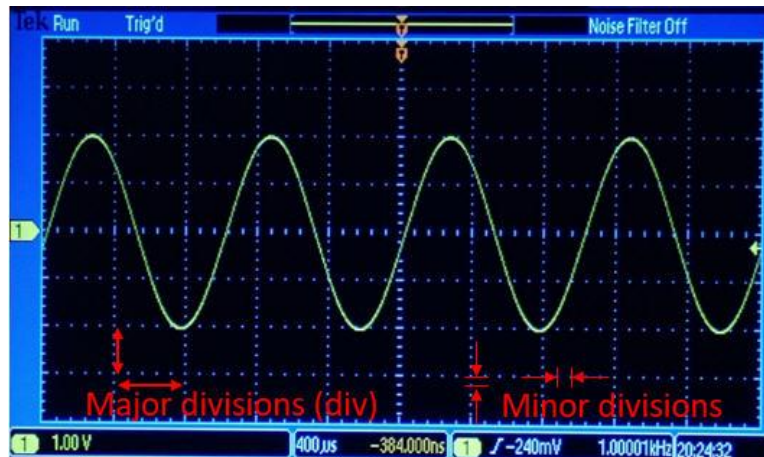


Figure 12. Example of 1 kHz signal

Trigger Control

- Now that you already have a nice signal on the screen, let's play with trigger control. explore what it actually does. The picture below shows the Figure 12 waveform with the oscilloscope trigger menu visible. **The trigger level in Figure 13 is indicated by a yellow arrow on right side of the scope face.**

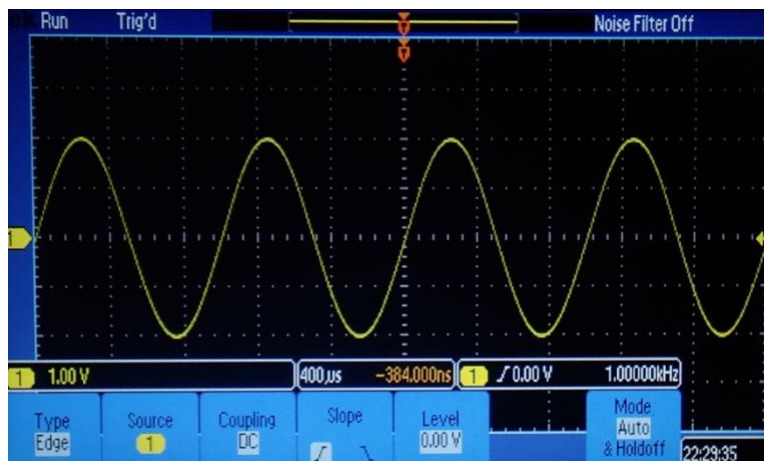


Figure 13. The same waveform as Figure 12 but with visible trigger menu.

- Switch the trigger slope to negative (falling edge) using the Trigger menu. What changes in the displayed signal? **Draw the signal in the scope face above or take a picture,** and label it clearly. Can you **deduce and draw where the trigger level is** from this signal and previous one? Explain.

- Turn the trigger level up as viewed on the scope. What changes in the displayed signal? **Draw the signal in the above signal display area or take a picture, and label it clearly.** Can you **deduce and draw where the new trigger level is?** Play with the trigger level to understand the correspondence between the trigger level and the displayed signal. **Explain what happens when the trigger level is tuned too high:**

- Next, switch from **Auto** to **Normal** mode using the displayed right sub-menu and buttons. Play with the trigger level. Is the displayed signal different from **Auto** mode when it is triggered? How about when it is **not** triggered? What does the “**Trig’d**” indicator in the scope face upper left tell you? **Answer below:**

when youre above it it doesn't move
switches to trigd? which means it is not on the wave

Vertical and horizontal control

- Set the trigger to **Auto**. Adjust the vertical and horizontal positions of the signal to make it centered.
- Change volts/div from 1 to 2. Does V_{pp} change? Count the divisions and calculate V_{pp} .

no

$$\underline{\quad 2 \quad} \text{ div} \times 2 \text{ volts/div} = \underline{\quad \quad} V_{pp}$$

- Change sec/div from $400\mu\text{s}$ to $200\mu\text{s}$. Does the period change? Count the divisions and calculate the period.

no

$$\underline{\quad 5 \quad} \text{ div} \times 200\mu\text{s/div} = \underline{\quad \quad} \text{ms}$$

Part 1: Measuring the Frequency-Dependence of a Low-Pass RC Filter

Theory

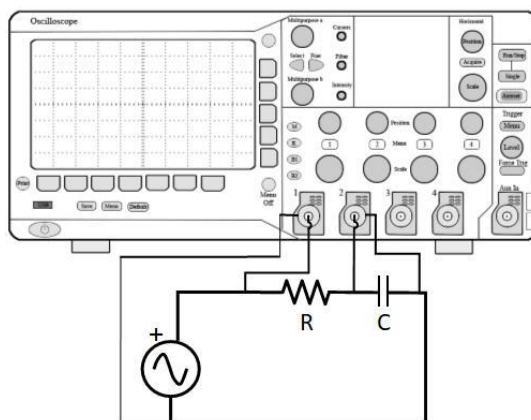


Figure 14. Hooking an RC Circuit to the Oscilloscope.

By connecting a capacitor in series to a resistor and applying an AC voltage, $V_0 \cos(\omega t)$, we can measure the frequency response of the RC circuit. The (real) peak to peak voltage $V_C(t)$ across the capacitor is given by equation (2a) as:

$$V_C(t) = \frac{V_0}{(1 + \omega^2 R^2 C^2)^{1/2}} \cos(\omega t + \phi_C) = V_C \cos(\omega t + \phi_C),$$

where V_C is the voltage amplitude. The phase difference ϕ_C is given by equation (2b):

$$\phi_C = \tan^{-1}(-\omega RC)$$

where the phase is determined by measuring the capacitor voltage waveform relative to the function generator voltage waveform in time.

The circuit in Figure 14 is called a “low-pass” filter because it transmits low frequencies and blocks high frequencies. The **break** or **cutoff frequency** ω_c is an important characteristic frequency of this filter, defined when $\omega_c = 1/RC$. For large frequencies $\omega \gg \omega_c$, the capacitor voltage approximates to:

$$V_C(t) = \frac{1}{\omega RC} V_0 \cos(\omega t - \pi/2),$$

while for low frequencies $\omega \ll \omega_c$, it becomes:

$$V_C(t) = V_0 \cos(\omega t)$$

You’ll notice for $\omega \gg \omega_c$, the amplitude should show a $V_0/\omega RC$ behavior. You can interpret $V_C(t)$ in this regime as a current, V_0/R , driving the capacitor with impedance $1/i\omega C$.

Setup

Circuit: Wire the RC circuit with $1 \text{ k}\Omega \pm 1\%$ resistor and a $1 \mu\text{F} \pm 5\%$ capacitor on the breadboard.

Function Generator: For the function generator, use the “T” BNC connector to split the output signal of the function generator into two BNC cables. Connect one coaxial cable to CH1 of the oscilloscope and the other BNC cable’s end to a BNC-to-minigrabber adapter, which is used to provide AC power the breadboard circuit.

Oscilloscope: Connect the capacitor to CH2 of the oscilloscope using a BNC with a BNC-to-minigrabber adapter and monitor the voltage across the capacitor.

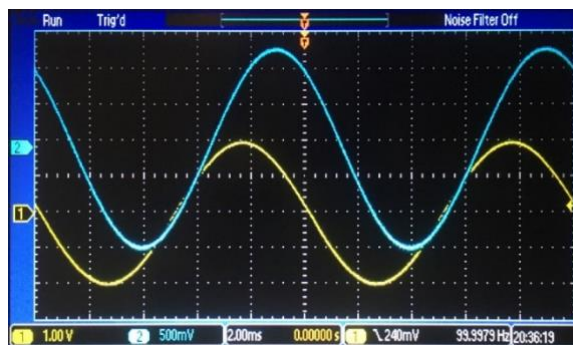


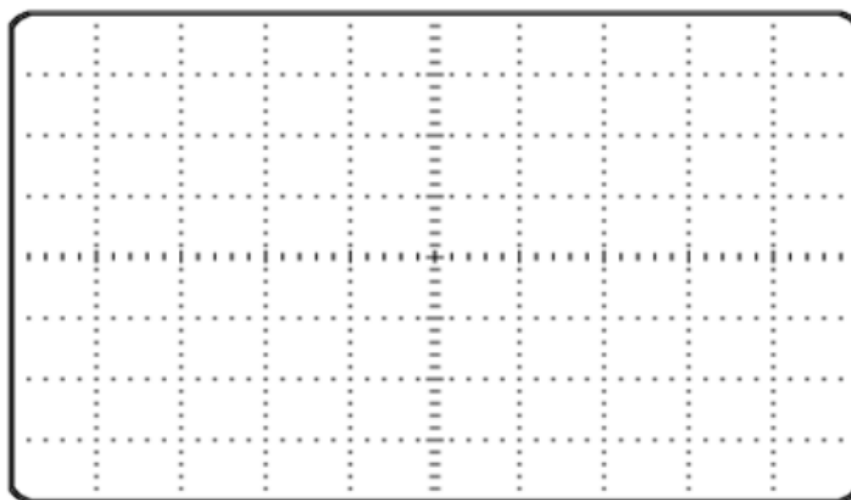
Figure 15. Example of RC circuit, capacitor signal with input signal set to @ $4 V_{pp}$, 100 Hz. The phase is negative because the capacitor signal occurs later in time than the function generator's input voltage.

Measure the Frequency-Dependent Voltages

Setup: Oscilloscope settings: set volts/div to be an appropriate value, and the coupling mode switch to DC for CH1 and CH2. Make sure you press the “2” button to view the signal from CH2 channel.

Setup: Set the function generator output signal to $4 V_{pp}$ 100 Hz sine wave. An example of possible signals on CH1 (yellow) and CH2 (blue) are shown in Figure 15.

Record: Now, draw your own waveforms in the scope face below, or take a picture. Fill in the table below:



Measure: Fill in the blank values from your signals:

$$\text{CH1: } \underline{\hspace{1cm}} \text{ div} \times \underline{\hspace{1cm}} \text{ volts/div} = \underline{\hspace{1cm}} V_{pp},$$

$$\text{CH1: } \underline{\hspace{1cm}} \text{ div} \times \underline{\hspace{1cm}} \text{ sec/div} = \underline{\hspace{1cm}} \text{ sec},$$

$$\text{CH2: } \underline{\hspace{1cm}} \text{ div} \times \underline{\hspace{1cm}} \text{ volts/div} = \underline{\hspace{1cm}} V_{pp},$$

$$\text{CH2: } \underline{\hspace{1cm}} \text{ div} \times \underline{\hspace{1cm}} \text{ sec/div} = \underline{\hspace{1cm}} \text{ sec},$$

$$\text{Phase: } \frac{\left(\underline{\hspace{1cm}} \text{ div} \times \frac{\text{msec}}{\text{div}} \right) \cdot 360^\circ}{\text{Signal period in msec}} = \underline{\hspace{1cm}}^\circ.$$

Note: The phase shift is negative if CH2 lies to the right (later time) of CH1.

Classroom check: Now, let us explore the frequency-dependent impedance. First, we'll need to define our frequency range, which is based on the cutoff frequency, ω_c . What is the expected cutoff frequency, f_c^{RC} , calculated from $\omega_c RC = 1$?

Referring to Figure 2, it should be clear that frequencies well below f_c^{RC} and well above f_c^{RC} supply important information on the circuit response. Frequencies where both the amplitude and phase change rapidly around f_c^{RC} define the shape. Finally, semi-log or log-log plots are used to graph the data, so frequencies in multiples of 1, 2 and 5 provide reasonable graph data spacing.

Data Collection: Select 7 different frequencies within the range 10 Hz and 10 kHz. Measure the peak-to-peak voltages and phase shift (you design this procedure), filling in the table below:

Table 1. Frequency dependence of capacitor voltage amplitude and phase

Frequency (Hz)	V_0 (Volts)	V_C (Volts)	Phase shift ϕ_C , (msec)	Phase shift ϕ_C (degrees)
Set on Freq. Generator	Measured (CH1)	Measured (CH2)	Measured (between CH signals)	Calculated (from prev column)
10				
100				
300				
700				
2500				
5000				
10 000				

Data Analysis - Plots

- Plot $\frac{V_C}{V_0}$ as a function of angular frequency, using a log-log scale.
- Fit the $\frac{V_C}{V_0}$ data with $\frac{V_C}{V_0} = \frac{A}{(1+\omega^2\tau^2)^{1/2}}$, where A is an arbitrary amplitude fit constant and $\tau = RC$ is a second fit constant, called the **time constant**.
[Note: This is Equation (2a), reformatted for fitting].
 - Solve for A , τ , and their errors.
 - Include your fit on the graph, and the fit parameter values.
 - If you are using Excel, you cannot calculate the A and τ errors. One of your group members should calculate in Python.
- Plot the phase shift ϕ_C as a function of angular frequency, using a semi-log scale. Include a theoretical phase shift line on the graph, using equation (2b). Obtain τ and include its value with uncertainty in the graph.
- Attach both the $\frac{V_C}{V_0}$ and the ϕ_C graphs with fit lines and fit parameters.**

Data Analysis - Questions

- For a series RC circuit, the time constant is expected to be $\tau = RC$. Does the τ calculated from the circuit elements values agree with the τ obtained from the fit? Explain, including the results from an acceptance test.
- Is the $\frac{V_C}{V_0}$ fit good at all frequencies? Write down possible reasons if that is not the case.
- Is the ϕ_C fit good at all frequencies? Write down possible reasons if that is not the case.

Part 2: Measuring the Frequency-Dependence of a High-Pass RL Filter

Theory

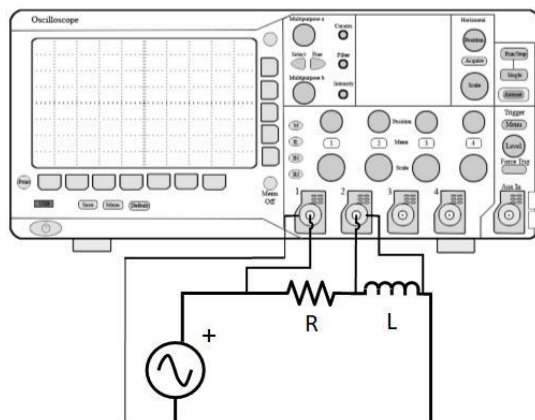


Figure 16. Hooking an RL Circuit to the Oscilloscope.

By connecting an inductor in series to a resistor and applying an AC voltage, $V_0 \cos(\omega t)$, we can measure the frequency response of the RL circuit. The (real) peak to peak voltage $V_L(t)$ across the capacitor is given by equation (3a) as:

$$V_L(t) = \frac{V_0 \omega L}{(R^2 + \omega^2 L^2)^{1/2}} \cos(\omega t + \phi_L) = V_L \cos(\omega t + \phi_L),$$

where V_L is the voltage amplitude. The phase difference ϕ_C is given by equation (3b):

$$\phi_C = \tan^{-1}\left(\frac{R}{\omega L}\right),$$

where the phase is determined by measuring the inductor voltage waveform relative to the function generator voltage waveform in time.

The circuit in Figure 16 is called a “high-pass” filter because it transmits high frequencies and blocks low frequencies. The cutoff frequency ω_c this filter is defined when $\omega_c = R/L$. For small frequencies $\omega \ll \omega_c$, the inductor voltage approximates to:

$$V_L(t) = \frac{\omega L}{R} \cdot V_0 \cos(\omega t - \pi/2),$$

while for low frequencies $\omega \gg \omega_c$, it becomes:

$$V_L(t) = V_0 \cos(\omega t)$$

You’ll notice for $\omega \ll \omega_c$, the amplitude should show an $\omega L V_0 / R$ behavior. You can interpret $V_L(t)$ in this regime as a current, V_0 / R , driving the inductor with impedance $i\omega L$.

Setup

Circuit: Wire the RL circuit with $10 \text{ k}\Omega \pm 1\%$ resistor and an $100 \text{ mH} \pm 10\%$ inductor on the breadboard.

Function Generator: Connect the function generator as in the previous part.

Oscilloscope: Connect the inductor to CH2 of the oscilloscope.

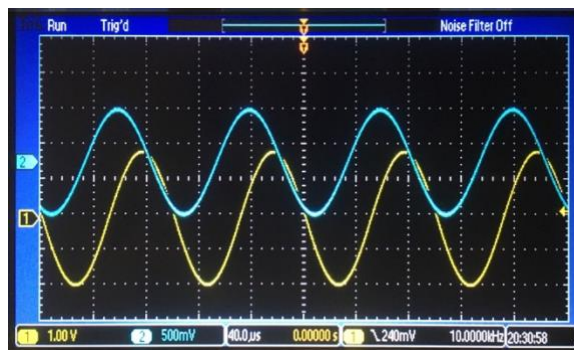


Figure 17. Example of RL circuit signal on oscilloscope screen. The phase is positive because the capacitor signal occurs later in time than the function generator's input voltage.

Measure the Frequency-Dependent Voltage and Impedance of an Inductor

Setup: Set volts/div of CH1 and CH2 to appropriate values.

Setup: Set the function generator output signal to 4 V_{pp} 10 kHz sine wave. An example of possible signals on CH1 (yellow) and CH2 (blue) are shown in Figure 17.

Note: The phase shift is positive if CH2 lies to the left (earlier time) of CH1.

Classroom check: Now, let us explore the frequency-dependent impedance. First, we'll need to define our frequency range, which is based on the cutoff frequency, ω_c . What is the expected cutoff frequency, f_c^{RL} ?

Referring to Figure 3, it should be clear that frequencies well below f_c^{RL} and well above f_c^{RL} supply important information on the circuit response. Frequencies where both the amplitude and phase change rapidly around f_c^{RL} define the shape. Finally, semi-log or log-log plots are used to graph the data, so frequencies in multiples of 1, 2 and 5 provide reasonable graph data spacing.

Warning: Inductors do not behave ideally in the real world, unlike capacitors!
You will further explore during analysis.

Data Collection: Select 7 different frequencies within the range 10 Hz and 50 kHz. Measure the peak-to-peak voltages and phase shift relative to the signal generator, filling in the table below.

Advanced: Consider using signal averaging and the Measure Amplitude and Phase selections instead of the cursors.

Table 2. Frequency dependence of inductor voltage amplitude and phase

Frequency (Hz)	V_0 (Volts)	V_C (Volts)	Phase shift ϕ_L , (msec)	Phase shift ϕ_L (degrees)
Set on Freq. Generator	Measured (CH1)	Measured (CH2)	Measured (between CH signals)	Calculated (from prev column)
10				
100				
500				
1 000				
5 000				
10 000				
50 000				

Data Analysis - Plots

- Plot $\frac{V_L}{V_0}$ as a function of angular frequency, using a log-log scale.
- Fit the $\frac{V_L}{V_0}$ data with $\frac{V_L}{V_0} = \frac{A\omega\tau}{(1+\omega^2\tau^2)^{1/2}}$, where A is an arbitrary amplitude fit constant and $\tau = L/R$ is a second fit constant.
 - Solve for A , τ , and their errors.
 - Include your fit on the graph, and the fit parameter values.
- Plot the phase shift ϕ_L as a function of angular frequency, using a semi-log scale. Include a theoretical phase shift line on the graph, using equation (3b) and the τ obtained from the $\frac{V_L}{V_0}$ fit.
- Attach both the $\frac{V_L}{V_0}$ and the ϕ_L graphs with fit lines and fit parameters.**

Data Analysis - Questions

For these data analysis questions remember that all inductors are not perfect! The measured inductor resistance for our inductor is $\sim 73 \Omega$.

- For a series RL circuit, the time constant is expected to be $\tau = L/R$. Does the τ calculated from the circuit elements' values agree with the τ obtained from the fit? Explain, using an acceptance test.
- Is the $\frac{V_L}{V_0}$ fit good at all frequencies? Write down possible reasons if that is not the case.
- Is the ϕ_L fit good at all frequencies? Write down possible reasons if that is not the case.
- Optional:** Can you propose an inductor model to explain the phase shift pattern? [Hint: Look in the literature]

Advanced Oscilloscope Operation

The “Acquire” button is used to average multiple traces. This measurement technique is useful when the signal is noisy or very low level. Press the Acquire button, then select the desired specifics using the scope menu buttons and multipurpose knobs.

The “Measure” button adds extensive real-time analytical capability while acquiring data. For Experiment 2, the “Amplitude” and “Phase” functions can be added. Multipurpose A and B knobs are used for selecting the appropriate inputs. An example of setting the 2->1 phase, is shown in Figure A3.

Waveforms can be saved to a USB drive using the Save/Recall buttons. Select the desired options (reduced number of points, csv format, source, etc.).

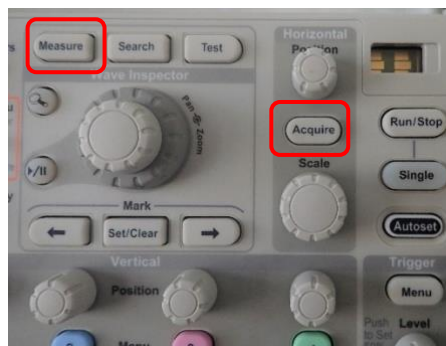


Figure A1. Acquire and Measure button locations on the oscilloscope

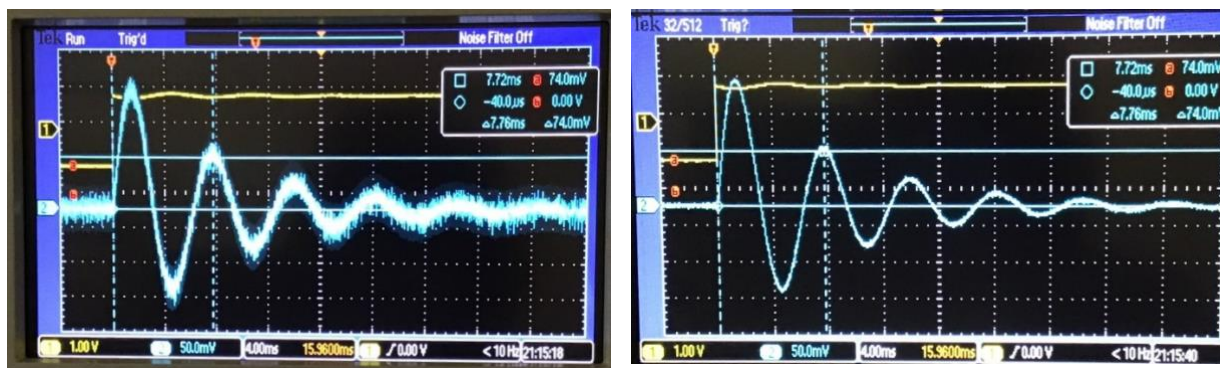


Figure A2. Improvement of Signal/Noise. Right trace is averaged 512 using Acquire, left has no averaging.

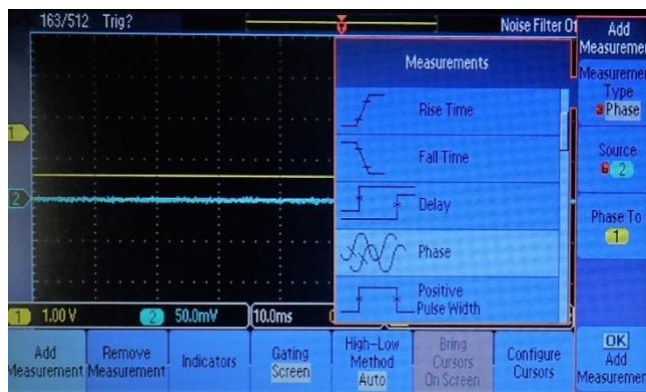


Figure A3. Setting the Measure Phase function.